

Slide 1: Human-Computer Interaction (HCI) in Minecraft

Focus: Visual affordances, targeting feedback, and direct inventory control.

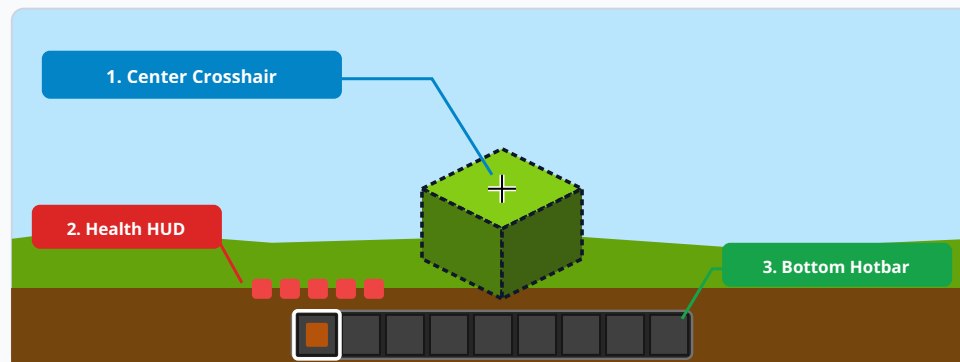


Figure 1: Screenshot annotation of user interface and direct manipulation controls.

3 HCI Interaction Elements & Usability Purpose

1. Center Reticle / Crosshair

Description & Role: A permanent targeting crosshair fixed in the center of the first-person display.

How it helps the user: Eliminates guesswork in 3D perspective by showing the exact point of interaction. When aiming at a block within reach, it renders a bounding outline around the cube, giving immediate visual confirmation before clicking.

2. Heads-Up Display (Health & Hunger Bars)

Description & Role: Icon rows positioned directly above the inventory hotbar.

How it helps the user: Communicates survival status at a glance without interrupting gameplay or forcing the player into nested menu screens. Flash and wobble animations provide instant feedback upon taking damage.

3. Bottom Item Hotbar & Active Selection Frame

Description & Role: A 9-slot horizontal tray showing equipped blocks and tools with a bright white selector box.

How it helps the user: Supports direct manipulation by binding items directly to keyboard numbers 1–9 and mouse scroll. Players switch held items instantaneously during fast-paced situations.

Slide 2: Computer Graphics (CG) & Spatial Analysis

Focus: 3D geometry rendering, lighting models, depth cues, and reflection.

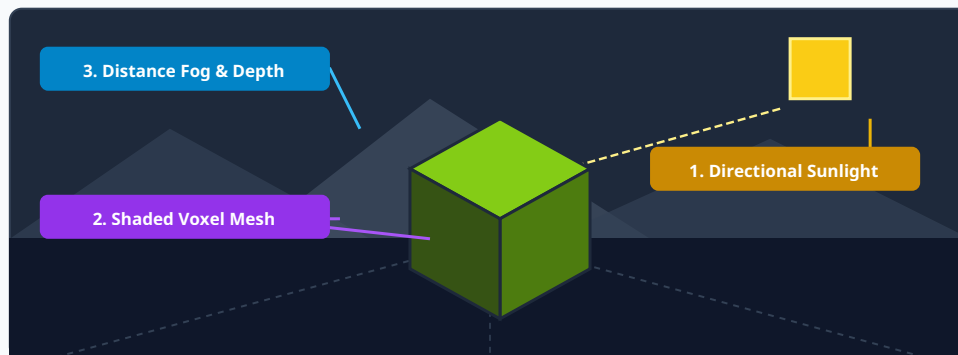


Figure 2: Rendering deconstruction showing directional illumination, polygon normals, and perspective.

3 Computer Graphics Elements & Spatial Creation

1. Directional Sunlight & Face Shading

Role in 3D Space: A global directional light source illuminates polygon faces based on their surface normal angles. The high contrast between the bright top face and dark side faces transforms 2D pixels into a perceived solid object.

2. Voxel 3D Mesh Geometry

Role in 3D Space: World geometry is composed of 3D polygonal cube meshes arranged on a discrete 3D coordinate grid (X, Y, Z). Each face carries coordinates, texture maps, and outward-facing normal vectors that define volume.

3. Perspective Camera Projection & Distance Fog

Role in 3D Space: Uses a perspective camera frustum where parallel lines converge toward the horizon, creating natural foreshortening. Distant terrain blends gradually into the sky via distance fog, creating atmospheric depth cues.

STEP 3: REFLECTION QUESTIONS

1. How does the Computer Graphics quality help or hinder interaction?

Minecraft's clean blocky geometry and distinct surface shading help interaction by making spatial distances and boundaries unambiguous without visual clutter. However, the lack of continuous surface curves and soft ambient shadows can occasionally make it difficult to judge exact vertical drops when looking straight down sheer cliff faces.

2. What is one simple HCI change you would make to improve the user interface?

Add an optional numerical percentage or durability counter directly over held tool icons in the bottom hotbar, allowing players to immediately check remaining tool life without memorizing hidden debug shortcuts (F3 + H).